

# Stochastic programming

## Lagrangian methods

Fabian Bastin

`bastin@iro.umontreal.ca`

Université de Montréal – CIRRELT – IVADO – Fin-ML

# Motivation

**Reference:** Birge and Louveaux, Sections 3.4, 5.8 and 6.4.

We consider the general **two-stage nonlinear program**

$$\begin{aligned} \inf z &= f^1(x) + Q(x) \\ \text{s.t. } g_i^1(x) &\leq 0, \quad i = 1, \dots, m_1, \end{aligned}$$

where  $Q(x) = E_\xi[Q(x, \xi)]$ , and

$$\begin{aligned} Q(x, \xi) &= \inf f^2(y(\xi), \xi) \\ \text{t.q. } b_i^2(x, \xi) + g_i^2(y(\xi), \xi) &\leq 0, \quad i = 1, \dots, m_2. \end{aligned}$$

We assume that all functions  $f^2(\cdot, \xi)$ ,  $b_i^2(\cdot, \xi)$  et  $g_i^2(\cdot, \xi)$  are continuous in the first argument for any given  $\xi$ , and measurable in  $\xi$  for any fixed first argument.  $\xi$  has finite second-order moments.

## Properties - assumptions

We can extend the definitions of  $K_1$  and  $K_2$ .

$$K_1 = \{x \mid g_i^1(x) \leq 0, i = 1, \dots, m_1\},$$

$$K_2(\xi) = \{x \mid \exists y(\xi) \text{ t.q. } b_i^2(x, \xi) + g_i^2(y(\xi), \xi) \leq 0, i = 1, \dots, m_2\},$$

$$K_2 = \{x \mid \mathcal{Q}(x) < \infty\}.$$

### Assumptions.

1. **Convexity.** The function  $f^1$  is convex on  $\mathbb{R}^{n_1}$ ,  $g_i^1$  is convex on  $\mathbb{R}^{n_1}$  ( $i = 1, \dots, m_1$ ),  $f^2(\cdot, \xi)$  is convex on  $\mathbb{R}^{n_2}$  for all  $\xi \in \Xi$ ,  $g_i^2(\cdot, \xi)$  is convex on  $\mathbb{R}^{n_2}$  ( $i = 1, \dots, m_2$ ), for all  $\xi \in \Xi$ , and  $b_i^2(\cdot, \xi)$  is convex on  $\mathbb{R}^{n_1}$  ( $i = 1, \dots, m_2$ ), for all  $\xi \in \Xi$ .
2. **Slater condition** If  $\mathcal{Q}(x) < \infty$ , for almost every (a.e.)  $\xi \in \Xi$ ,  $\exists$  some  $y(\xi)$  s.t.  $b_i^2(x, \xi) + g_i^2(y(\xi), \xi) < 0$  ( $i = 1, \dots, m_2$ ).

## Second-stage properties

The Slater condition ensures that the *strong duality* holds at the second stage and the KKT conditions are necessary and sufficient.

### Theorem

*Under assumptions 1 and 2, the recourse function  $Q(x, \xi)$  is a convex function in  $x$  for all  $\xi \in \Xi$ .*

### Theorem

*If the recourse feasible set (i.e. in  $y$ , for given  $\xi$ ) is bounded for any  $x \in \mathbb{R}^{n_1}$ , then the function  $Q(x, \xi)$  is lower semi-continuous in  $x$  for all  $\xi \in \Xi$ .*

## Lower semi-continuity

We say that a function  $f$  is **lower semi-continuous in  $x_0$**  if for every  $\varepsilon > 0$ ,  $\exists$  a neighborhood  $U$  of  $x_0$  such that  $f(x) > f(x_0) - \varepsilon$  for all  $x \in U$ . Equivalently

$$\liminf_{x \rightarrow x_0} f(x) \geq f(x_0),$$

where

$$\liminf_{x \rightarrow a} f(x) = \lim_{\varepsilon \rightarrow 0} \inf \{ f(x) : x \in \text{dom}(f) \cap B(a; \varepsilon) \setminus \{a\} \}.$$

The function  $f$  is **lower semi-continuous** if it is lower semi-continuous in any point in its domain.

## Lower semi-continuity

A function is lower semi-continuous iff  $\{x \in X : f(x) > \alpha\}$  is an open set for every  $\alpha \in \mathbb{R}$ , or, in a similar way,  $\{x \in X : f(x) \leq \alpha\}$  is a closed set for every  $\alpha \in \mathbb{R}$ .

Example of lower semi-continuous function

$$f(x) = \begin{cases} x & \text{if } x \leq 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

## Convexity (cont'd)

### Corollary

*The expected recourse function  $Q(x)$  is a convex function in  $x$ .*

### Corollary

*The feasible set  $K_2 = \{x \mid Q(x) < \infty\}$  is closed and convex.*

### Corollary

*If the recourse feasible set is bounded for any  $x \in \mathbb{R}^{n_1}$ ,  $Q$  is a lower semi-continuous function.*

## Solution: existence

### Theorem

*If the recourse feasible set is bounded for any  $x \in \mathbb{R}^{n_1}$ ,  $K_1$  is bounded,  $f^1$  is continuous,  $g_i^1$  and  $g_i^2$  are continuous for every  $i$ , and  $K_1 \cap K_2 \neq \emptyset$ , then the nonlinear stochastic two-stage program has an optimal solution and the infimum is reached.*



## Solution: optimality

### Theorem

*Assume that the Slater condition is satisfied:*

*$\exists x$  s.t.  $x \in \text{ri}(\text{dom}(f^1(x))), x \in \text{ri}(\text{dom}(\mathcal{Q}(x)))$  and  $g_i^1(x) < 0, i = 1, \dots$*

*$x^*$  is optimal iff there exists  $\lambda^*$  such that  $(x^*, \lambda^*)$  satisfies the KKT conditions for the two-stage stochastic program, i.e.*

- $x^* \in K_1,$
- $\lambda^* \geq 0,$
- $\lambda_i^* g_i^1(x^*) = 0, i = 1, \dots, m,$
- $0 \in \partial f^1(x^*) + \partial \mathcal{Q}(x^*) + \sum_{i=1}^{m_1} \lambda_i^* \partial g_i^1(x^*).$

## Relative interior

The **relative interior** of a set  $S$ , denoted by  $ri(S)$ , is defined as its interior within the affine envelop of  $S$ . In other words

$$ri(S) = \{x \in S \mid \exists \varepsilon > 0, (B_\varepsilon(x) \cap \text{aff}(S)) \subseteq S\},$$

where  $\text{aff}(S)$  is the **affine envelop** of  $S$ , and  $B_\varepsilon(x)$  is a ball of radius  $\varepsilon$  centered at  $x$ . Any metric can be used for the ball construction: all define the same relative interior.

The **affine envelop**  $\text{aff}(S)$  of  $S$  is the set of all affine combinations of elements of  $S$ , i.e.

$$\text{aff}(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid x_i \in S, \alpha_i \in \mathbb{R}, \sum_{i=1}^k \alpha_i = 1, k = 1, 2, \dots \right\}.$$

# Basic principle of Lagrangian approach

- Nonlinear problem.
- We aim to build a search direction and compute a step along this direction to reduce the objective.
- Problem: classical nonlinear methods assume that the (sub-)gradients of  $\mathcal{Q}$  are available and cheap to obtain. Not the case here.
- Link the first and second stages in order to avoid optimization subproblems when building search directions.

## Basic principle of Lagrangian approach (cont'd)

- Let  $\pi$  be a vector of (dual) multipliers associated to the second-stage constraints.
- Dual problem with respect to the second-stage constraints (linking the first and second stages):

$$\max_{\pi(\xi) \geq 0} w = \theta(\pi),$$

where

$$\begin{aligned} \theta(\pi) = & \inf_{x, y} f^1(x) + \mathbb{E}_{\xi}[f^2(y(\xi), \xi)] + \\ & \mathbb{E}_{\xi} \left[ \sum_{i=1}^{m_2} \pi_i(\xi) \left( b_i^2(x, \xi) + g_i^2(y(\xi), \xi) \right) \right] \\ & \text{s.t. } g_i(x) \leq 0, i = 1, \dots, m_1. \end{aligned}$$

# Duality

## Theorem

Assume that

- all the functions of the stochastic program are convex,
- there exists a finite optimal value,
- the Slater constraints qualification condition holds,
- the support of  $\xi$  is finite.

Then  $z \geq w$  for all  $x, y_1, \dots, y_S$  feasible in the primal formulation and  $\pi_1, \dots, \pi_S$  feasible in the dual formulation (weak duality).

Moreover, their optimal values coincide:  $z^* = w^*$  (strong duality).

We thus assume from now that  $\Xi$  is finite, and we will describe some procedures exploiting the Lagrangian function.

# Basic Lagrangian dual ascent method

Assumption: the dual problem always has a unique solution.

## Algorithm 1: Lagrangian dual ascent method

**Step 0.** Let  $\pi^{0,s} \geq 0$ ,  $s = 1, \dots, S$ ,  $\pi^0 = (\pi^{0,1}, \dots, \pi^{0,S})$ ,  $\nu = 0$ .

**Step 1.** Given  $\pi = \pi^\nu$  in the dual problem, consider the solution  $(x^\nu, y_1^\nu, \dots, y_S^\nu)$ . For  $s = 1, \dots, S$ , define

$$\hat{\pi}_i^s = b_i^2(x^\nu, s) + g_i^2(y_s^\nu, s).$$

If  $\hat{\pi}^s = 0$  for all  $s$ , stop.

**Step 2.** Choose a step  $\alpha^\nu \geq 0$  and set  $\pi^{\nu+1,s} = \max\{\pi^{\nu,s} + \alpha^\nu \hat{\pi}^s, 0\}$ . Set  $\nu := \nu + 1$  and check convergence. If we have not yet converged, return to Step 1.

## Properties - convergence

- Various strategies can be used to compute the step  $\alpha^\nu$ . Assume we compute the step to maximize  $\theta(\pi^\nu + \alpha\hat{\pi})$  on  $\alpha \geq 0$ . Under the assumption of unicity of dual solution, we can show that this algorithm always produces an **ascent direction** in  $\theta$ .
- Either the algorithm converges to an optimal solution, or, assuming a bounded set of optimal solutions, it produces an infinite sequence where all limit points are optimal.
- Good performance if the number of dual iterations is small compared to the number of function evaluations that would be required by solving the original problem directly.
- Solving the dual program may be time-consuming, but should nevertheless be easier than solving the original problem directly, as the constraints linking the two stages (i.e. involving  $x$  and  $y(\xi)$ ) now appear in the objective.

## Another approach: scenarios

As  $\Xi$  is finite, we can consider that  $S$  scenarios exist, and we can link a non-anticipative decision  $\hat{x}$  and the decisions of the scenario  $s$  with the terms  $(\hat{x} - x_k)$  in the objective function.

Consider the general problem

$$\inf_{x \in \mathcal{N}} \mathbb{E} \left[ \sum_{t=0}^T f_{t+1}(\xi, x^t(\xi), x_{t+1}(\xi)) \right],$$

s.t.  $x_t(\xi) \in X_t(\xi)$  a.s.,

where  $x_t$  is the decision at stage  $t$ , and  $x^t$  is the decisions history at stage  $t$ , i.e.  $x^t = \{x_1, \dots, x_t\}$ .  $\mathcal{N}$  designs the closed linear subspace of nonanticipative processes. Denote  $\mathcal{A}$  the  $\sigma$ -field of all the events (i.e. the collection of events subsets, with the probability measure associated to  $\xi$ ).



# Nonanticipativity

Mathematically,

- the nonanticipativity aims to require that  $x^t(\xi)$  has to be  $\mathcal{A}^t$ -measurable, where  $\mathcal{A}^t$  is the  $\sigma$ -field of events until time  $t$ ;
- or, in other words,  $x^t(\xi) = E[x^t(\xi) | \mathcal{A}^t]$  a.s.,  $t = 0, \dots$ ;
- using the projection operator  $\Pi^t : z \rightarrow \Pi^t z := E[z | \mathcal{A}^t]$ , this is equivalent to

$$(I - \Pi^t)x^T = 0, \quad t = 0, \dots$$

## Nonanticipativity (cont'd)

Let  $S_s^t$  be the set of scenarios identical to the scenario  $s$  at time  $t$ . We then have to ensure that

$$x_{its} = x_{its'}, \forall i \in N, \forall t \in T, \forall s \in S, \forall s' \in S_s^t.$$

**Idea:** explicitly include the nonanticipativity constraints, and place them in the objective function, rather than in the second stage constraints.

## Dual ascent and scenarios

Consider the maximization of the dual built on the nonanticipativity constraints:

$$\max_{\pi(\xi) \geq 0} w = \theta(\pi),$$

where

$$\begin{aligned} \theta(\pi) = \inf_{x \in \mathcal{X}} z = & \mathbb{E} \left[ \sum_{t=0}^T f_{t+1}(\xi, x^t(\xi), x_{t+1}(\xi)) \right] \\ & + \mathbb{E} \left[ \sum_{t=0}^T \pi^t(\xi) (I - \Pi_t) x^t(\xi) \right], \end{aligned}$$

where  $\pi^t$  corresponds to the components of  $\pi$  associated to the  $t$  first periods.

## Dual ascent and scenarios: two stages

We still have to describe the projection operator. For simplicity, consider two steps. The primal problem becomes (omitting all the constraints except the nonanticipativity ones)

$$\begin{aligned} \min z &= \sum_{s=1}^S p_s \left( f^1(x_s) + f^2(x_s, y_s) \right) \\ \text{s.t. } x_s - \sum_{k=1}^S p_k x_k &= 0, \quad s = 1, \dots, S. \end{aligned}$$

The dual problem can now be written as

$$\begin{aligned} \max_{\pi} \theta(\pi) &= \min_{x, y} \sum_{s=1}^S p_s \left( f^1(x_s) + f^2(x_s, y_s) \right) \\ &\quad + \pi_s \left( x_s - \sum_{k=1}^S p_k x_k \right). \end{aligned}$$

## Dual ascent and scenarios: algorithm

### Algorithm 2: Lagrangian dual ascent method

**Step 0.** Let  $\pi^0 \geq 0$ ,  $\nu = 0$ .

**Step 1.** Given  $\pi = \pi^\nu$  in the dual problem, compute the solution  $(x_1^\nu, \dots, x_S^\nu, y_1^\nu, \dots, y_S^\nu)$ .

**Step 2.** If  $x_s - \sum_{k=1}^S p_k x_k = 0$ ,  $s = 1, \dots, S$ , stop: the solution is optimal.  
Otherwise, define  $\hat{\pi}_s = x_s - \sum_{k=1}^S p_k x_k$ , and go to Step 3.

**Step 3.** Let  $\alpha^\nu$  maximize  $\theta(\pi^\nu + \alpha \hat{\pi})$  on  $\pi^\nu + \alpha \hat{\pi} \geq 0$ ,  $\alpha \geq 0$ . Set  $\pi^{\nu+1} = \pi^\nu + \alpha^\nu \hat{\pi}$ ,  $\nu = \nu + 1$ . Return to Step 1.

# Augmented Lagrangian

- Unfortunately, usually slow as there is a linearization at a single point of  $\theta$  only. It is nevertheless easy to implement and can give good results, especially for small size problems.
- In order to improve the performances, we will turn to another technique inspired by nonlinear programming: **augmented Lagrangian** approaches.
- Augmented Lagrangian: add a penalty on  $\theta(\pi)$ .
- Moreover, with augmented Lagrangian techniques, we can try to exploit the problem structure in order to decompose the problem.

# Augmented Lagrangian and Stochastic Programming

We again develop the ideas in the two-stages context, while the approach can be easily generalized in the multi-stage context.

Recall that we aim to solve

$$\max_{\pi(\xi) \geq 0} w = \theta(\pi),$$

where

$$\begin{aligned} \theta(\pi) = \inf_{x \in \mathcal{X}} z = & \mathbb{E} \left[ \sum_{t=0}^T f_{t+1}(\xi, x^t(\xi), x_{t+1}(\xi)) \right] \\ & + \mathbb{E} \left[ \sum_{t=0}^T \pi^t(\xi) (I - \Pi_t) x^t(\xi) \right], \end{aligned}$$

Penalizing the nonanticipativity constraints, we obtain the following program, where  $\hat{x}$  is nonanticipative.

## Augmented Lagrangian and SP: two stages

$$\begin{aligned}\theta(\pi) = \inf z = f^1(\hat{x}) + \sum_{s=1}^S \left( p_s f^2(y_s, \xi_s) + \pi_s^T (x_s - \hat{x}) + \frac{\rho}{2} \|x_s - \hat{x}\|^2 \right) \\ \text{s.t. } g_i^1(\hat{x}) \leq 0, \quad i = 1, \dots, m_1, \\ t_i^2(x_s, \xi_s) + g_i^2(y_s, \xi_s) \leq 0, \quad i = 1, \dots, m_2, \quad s = 1, \dots, S.\end{aligned}$$

$\rho$  is the augmented Lagrangian penalty parameter, and  $\pi_s$  is the vector of dual variables associated to the nonanticipativity constraints of the scenario  $s$ .

The method aims to contract the pair  $(\hat{x}^{\nu+1}, \pi^{\nu+1})$  around a saddle point.



## Progressive hedging

- Main reference: R. T. Rockafellar and R. J.-B. Wets, *Scenarios and policy aggregation in optimization under uncertainty*, Mathematics of Operations Research 16(1):119–147 (1991).  
The paper focussed on the multistage version.
- The method performs a complete separation of the problems, scenario by scenario, at each iteration, reducing the cost per iteration.
- The number of iterations can however increase. . .
- But for structured problems, it is possible to exploit parallelism, and therefore solve large-scale problems.
- **Observation**: the Lagrangian function is not separable with respect to the scenarios due to the term  $(\hat{x} - x_s)$ .

## Progressing hedging: basic principle

Alternatively,

1. fix  $\hat{x}$  and obtain the solutions  $x_s$ ,  $s = 1, \dots, S$ ,
2. fix  $x_s$ ,  $s = 1, \dots, S$ , and compute  $\hat{x}$ .

In other words, we work scenario by scenario, and we enforce progressively the nonanticipativity constraints.

At iteration  $\nu$ , we solve the subproblems

$$\begin{aligned} \inf z = & \sum_{s=1}^S p_s \left( f^1(x_s) + f^2(y_s, \xi_s) + (\pi_s^\nu)^T (x_s - \hat{x}^\nu) + \frac{\rho}{2} \|x_s - \hat{x}^\nu\|^2 \right) \\ \text{s.t. } & g_i^1(x_s) \leq 0, \quad i = 1, \dots, m_1, \quad s = 1, \dots, S, \\ & t_i^2(x_s, k) + g_i^2(y_s, \xi_s) \leq 0, \quad i = 1, \dots, m_2, \quad s = 1, \dots, S. \end{aligned}$$

$\hat{x}^\nu$  is not necessarily a feasible solution!

# Decomposition

However, it is easy to decompose the problem!

Algorithm:

**Step 0.** Assume that we have a nonanticipative solution  $x^0$ , an initial vector of multipliers  $\pi^0$ , and  $\rho > 0$ .  $\nu \leftarrow 0$ . Go to Step 1.

**Step 1.** Let  $(x_s^{\nu+1}, y_s^{\nu+1})$  be a solution of the previous Lagrangian program, for  $s = 1, \dots, S$ . Set

$$\hat{x}^{\nu+1} = \left( \hat{x}_1^{\nu+1}, \dots, \hat{x}_S^{\nu+1} \right)^T,$$

$$\text{where } \hat{x}_s^{\nu+1} = \sum_{s=1}^S p_s x_s^{\nu+1}, \quad s = 1, \dots, S.$$

## Decomposition (cont'd)

**Step 2.** Let  $\pi_s^{\nu+1} = \pi_s^\nu + \rho (x_s^{\nu+1} - \hat{x}_s^{\nu+1})$ ,  $s = 1, \dots, S$ .

If  $\hat{x}^{\nu+1} = \hat{x}^\nu$  and  $\pi^{\nu+1} = \pi^\nu$ , stop:  $(\hat{x}^\nu, \pi^\nu)$  is optimal.

Otherwise, set  $\nu = \nu + 1$  and return to Step 1.

- Step 2 simply consists to take the expected value of  $x^{\nu+1}$  as  $\hat{x}^{\nu+1}$ .
- The basis of this approach is therefore the contraction of the pair  $(\hat{x}^\nu, \pi^\nu)$  around a saddle point rather than a dual ascent strategy.
- In practice, the optimality test is replaced by a convergence test for instance if the following quantity is small enough (De Silva et Abramson):

$$\sqrt{\|\hat{x}^\nu - \hat{x}^{\nu+1}\|^2 + \sum_{s=1}^S \rho_s \|x_s^\nu - \hat{x}^\nu\|^2}.$$

## Generalization to multistage

$$\min_x \sum_{s \in \mathcal{S}} p_s f(x^{(s)}, \xi^{(s)})$$

$$\text{s.t. } x^{(s)} \in \mathcal{X}^{(s)},$$

$$x_t^{(s)} \text{ is nonanticipative, } t = 1, \dots, T.$$

Nonanticipativity:

$$x_t^{(s)} = \mathbb{E} \left[ x_t^{(s')} \mid s' \in \mathcal{S}_t^{(s)} \right],$$

where  $\mathcal{S}_t^{(s)} = \{s' \mid \bar{\xi}_t^{(s)} = \bar{\xi}_t^{(s')}\}$ . It can be reformulated as

$$x_t^{(s)} = \hat{x}_t^{(s)},$$

with

$$\hat{x}_t^{(s)} = \frac{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'} x_t^{(s')}}{\sum_{s' \in \mathcal{S}_t^{(s)}} p_{s'}}.$$

# Generalization

Augmented Lagrangian:

$$L(x, \pi, \rho) = \mathbb{E} \left[ f \left( x^{(s)}, \xi^{(s)} \right) + \sum_{t=1}^T \left( \pi_t^{(s)'} \left( x_t^{(s)} - \hat{x}_t^{(s)} \right) + \frac{\rho}{2} \left\| x_t^{(s)} - \hat{x}_t^{(s)} \right\|^2 \right) \right],$$

where  $\pi$  is the Lagrange multipliers vector associated to the NA constraints, and  $\rho > 0$  is a penalty parameter.

# Implementation

The `StochasticPrograms.jl` package proposes an implementation of the progressive hedging algorithm, but limited to two stages.

Some variations or unmaintained versions exist.

# Generalization

Given  $\pi$ , the augmented Lagrangian program is

$$\begin{aligned} \min_x \quad & L(x, \pi, \rho) \\ \text{s.t.} \quad & x^{(s)} \in \mathcal{X}^{(s)}, s \in \mathcal{S}. \end{aligned} \tag{1}$$

In order to achieve full separability, fix  $\hat{x}_t^{(s)}$  and repeatedly solve the program by updating the Lagrange multipliers vector and the value of  $\hat{x}_t^{(s)}$  between consecutive resolutions.



## Progressive hedging algorithm (PHA)

**Step 0.** Set  $\hat{x}^{(s),0} = (\hat{x}_1^{(s),0}, \dots, \hat{x}_T^{(s),0})$  and  $k = 0$ . Choose  $\pi^{(s),0} = \mathbf{0}$ ,  $\rho^0 > 0$ .

**Step 1.** Compute  $x^{(s),k+1} = (x_1^{s,k+1}, \dots, x_T^{s,k+1})$ ,  $s = 1, \dots, S$ , by solving each scenario subproblem

$$\begin{aligned} \min_{x^{(s)}} & f(x^{(s)}, \xi^{(s)}) + \sum_{t=1}^T \left( \pi_t^{(s)'} (x_t^{(s)} - \hat{x}_t^{(s),k}) \right. \\ & \left. + \frac{\rho^k}{2} \|x_t^{(s)} - \hat{x}_t^{(s),k}\|^2 \right) \\ \text{s.t. } & x^{(s)} \in \mathcal{X}^{(s)}. \end{aligned}$$

**Step 2.** For  $s = 1, \dots, S$ ,  $t = 1, \dots, T$ , set

$$\hat{x}_t^{(s),k+1} = \frac{\sum_{s' \in \mathcal{S}_t^{(s)}} \rho_{s'} x_t^{(s'),k+1}}{\sum_{s' \in \mathcal{S}_t^{(s)}} \rho_{s'}}.$$

# Progressive hedging algorithm (PHA)

Step 3. Set  $\rho^{k+1}$  and

$$\pi_t^{(s),k+1} = \pi_t^{(s),k} + \rho^k \left( x_t^{(s),k+1} - \hat{x}_t^{(s),k+1} \right),$$

for  $t = 1, \dots, T$ ,  $s \in \mathcal{S}$ .

Step 4. Stop if convergence is achieved. Otherwise, set  $k \leftarrow k + 1$  and return to Step 1.

# Convergence

## Theorem

*Assume that the initial nonlinear stochastic program has only convex functions and a finite optimal value, as well as a strictly feasible point. Assume moreover that the support of  $\xi$  is finite. The progressive hedging algorithm converges to an optimal solution  $x^*, \pi^*$ .*

- De Silva and Abramson have tested the progressive hedging algorithm for linear problems in portfolio management, proposed by Mulvey and Vladimirou.
- The subproblems were solved by means of interior points methods.
- Good speed-up convergence, but the algorithm remains sensitive to the choice of the penalty parameter.